

Implementation of a Panorama Stitching Pipeline with Geometric Alignment and Seamless Blending

Prashant Kumar
M25CSE023

1. Abstract

The objective of panorama stitching is to merge multiple overlapped images into a wide-field image such that geometric consistent and visually continuous are preserved. In this paper, a full panorama stitching pipeline is established with feature-based matching and manual image warping. I use SIFT to detect the key points and describe these features, then match them between two images and estimate a robust homography (DLT with RANSAC). The images warped to a common reference system and their associated sky backgrounds after exposure alignment and blending are presented for inspection in order to provide an estimate of the depth that can be reached using this mosaic technique. The comparison between an uncorrected stitched panorama and a corrected final blend allows one to appreciate the necessity of photometric correction in panoramic imaging.

2. Introduction

Panoramic Panorama imaging is a method that composes images from different perspective into a much broader view. These kinds of images are widely used in applications such as virtual tours, mapping, architectural visualization and immersive photography. However, producing a visually pleasing panorama is difficult because of geometrical inconsistencies, exposure discrepancies and visible artifacts in overlapping areas. Even if you take pictures carefully, tiny differences in lighting and exposure can still cause your photos to look different. Feature correspondences that are wrong could also cause geometric distortion. Thus, a successful panorama stitching pipeline should cater to not only geometric alignment but also photometric coherence. This work realizes a whole panorama stitching system based on simple image processing and linear transformations, without employing any high-level automatic stitching interface.

3. Objective

The objectives are:

- To stitch 3 overlapping images into a single panoramic image.
- To establish geometric alignment using feature-based correspondence.
- To estimate homography using DLT with RANSAC for robustness.
- To perform manual image warping using inverse mapping.
- To compare naive stitching results with a blended panorama.

4. Image Acquisition

Three Left, Center and Right images were taken with a mobile phone camera by rotating the camera while keeping it stationary in a fixed location to avoid parallax effects. An overlap of 30–40% between successive images was kept. The middle image was set to be the reference frame for panorama merging.

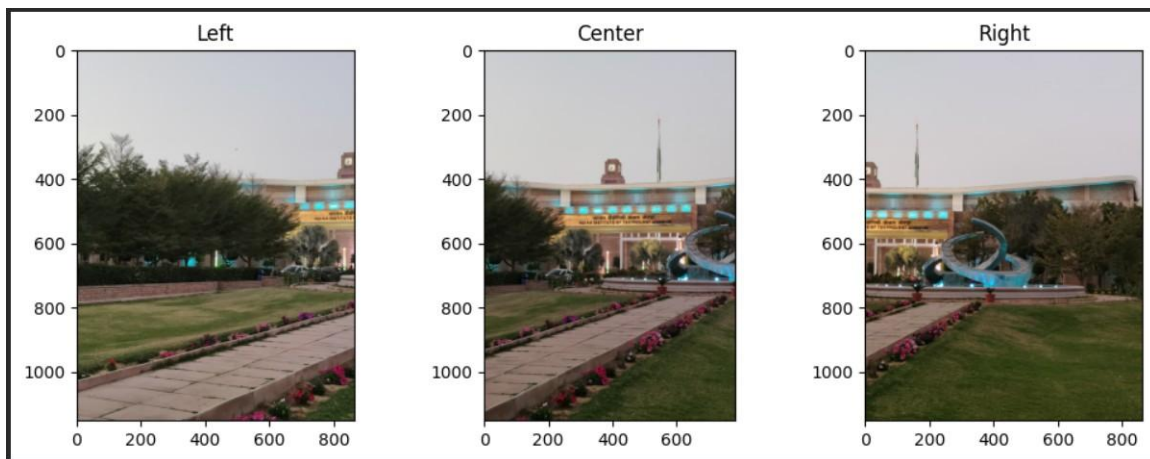


Figure 1: We take 3 images (Fig. left to right), with (30% - 40%) overlap to be stitched into a panorama.

5. Methodology

5.1 Extraction of Features

The scale-invariant feature transform (SIFT) is employed to find distinctive key-points and compute the descriptors for images. These are scale and rotate invariant features which can be used for matching overlapping images.

5.2 Feature Matching

A brute-force matcher is used to find all of the matches that occur between the left-center right-center pairs of images. Using the Lowe's ratio test, we remove the unreliable matches and keep only robust correspondences.

5.3 Computing the Homography

A homography is a 3×3 matrix describing the projective transformation between two image planes. This matrix is estimated by DLT algorithm from corresponding feature points and solved using Singular Value Decomposition (SVD) to establish the linear system.

5.4 Mathematical Model

The geometric relationship between two overlapping images is modeled using a planar homography. A homography maps a point from one image plane to another and is represented by a 3×3 matrix.

• Homography Mapping

A source image point $\mathbf{x} = (x, y, 1)^T$ maps to a dest. image point $\mathbf{x}' = (x', y', 1)^T$ by:

$$\mathbf{x}' \sim \mathbf{H}\mathbf{x} \quad (1)$$

here, $\mathbf{H}$ is the homography matrix and $\sim$ mean equal up to scale.

• DLT Constraint

For corresponding points (x, y) and (x', y') , the Direct Linear Transformation (DLT) satisfy constraint:

$$\begin{bmatrix} -x & -y & -1 & 0 & 0 & 0 & xx' & yx' & x' \\ 0 & 0 & 0 & -x & -y & -1 & xy' & yy' & y' \end{bmatrix} \mathbf{h} = 0 \quad (2)$$

where $\mathbf{h}$ is the vectorized homography matrix.

• RANSAC Reprojection Error

The homography quality is determined by the re-projection error RANSAC:

$$e = \|\mathbf{x}' - \hat{\mathbf{x}}'\|_2 \quad (3)$$

where $\hat{\mathbf{x}}'$ is the projected point using the estimated homography.

5.5 Robust Estimation via RANSAC

RANSAC is employed to deal with outliers in the matches of features. Homographies are computed from random subsets of correspondences, and the one with the largest cardinality among inliers is chosen as evening. Multiple homographies are estimated from randomly selected sub-sets of correspondence, and the one with the maximum number of inliers is called

final.

5.6 Image Warping

Images are transformed to a common coordinate system according to the computed homography. We used inverse warping to prevent hole in the output image and bilinear interpolation to calculate pixel values at non-integer coordinates.

5.7 Naive Panorama Construction

The naive panorama is created by putting the warped images on top of each other directly (no exposure compensation or blending). This is a minimum result.

5.8 Exposure Alignment

Because the images are exposed differently, brightness variations exist even on overlapping regions. Exposure alignment is done by taking the gain from computing the median brightness values of overlapping regions and applying it to normalize image intensities. The exposure correction factor is computed for all overlapping regions in their median luminances:

$$g = \frac{\text{median}(I_{\text{ref}})}{\text{median}(I_{\text{src}})} \quad (4)$$

where I_{ref} and I_{src} are the luminance values of the reference and source images, respectively. The gain factor g is used to multiply with the source image for brightness insensitivity.

5.9 Seamless Blending

Visible seams are eliminated by seamless blending achieved by means of weighted averaging in overlapping areas. This way there is smooth transition between images and it maintains the visual continuity.

5.10 Cropping and Boundary Refinement

Warping image onto a larger canvas introduces black region where valid pixels do not exist. Remove this region by cropping, resulting in rectangular panorama.

6. Results

6.1 Naive Panorama

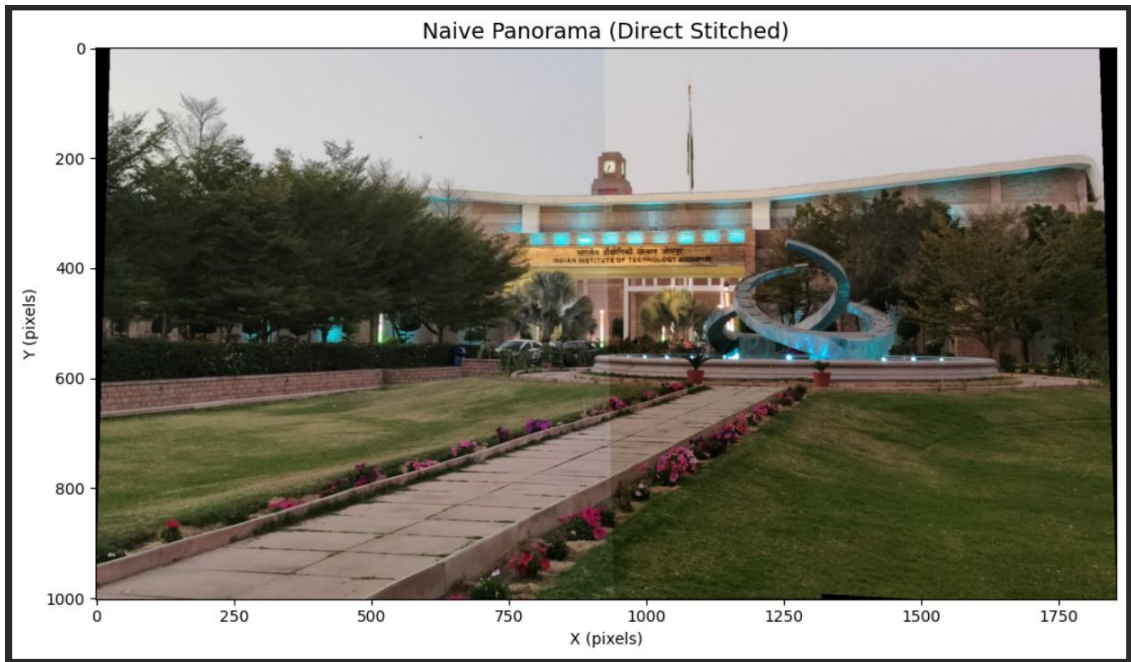


Figure 2: Demonstrates a naive panorama, obtained by direct stitching, with no blending or exposure correction. Here the seams are clearly visible as are the brightness differences.

6.2 Final Blended Panorama

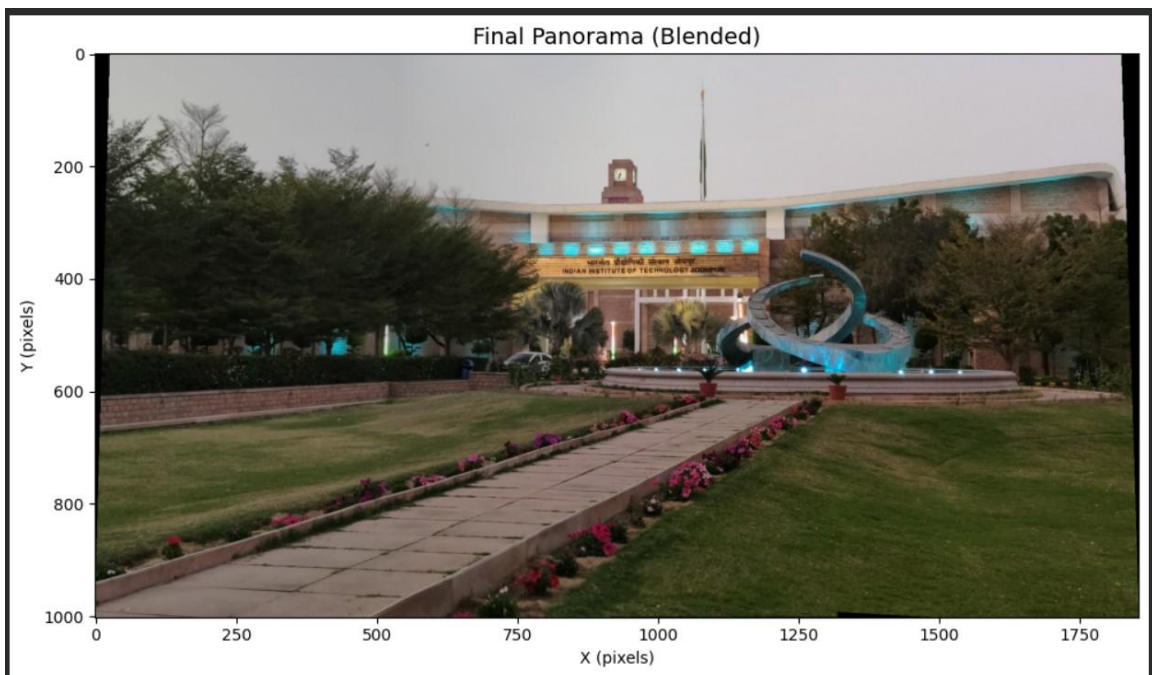


Figure 3: Final illustrates the final panorama after seamless blending.

7. Comparison and Discussion

The Comparison of naive panorama and touch-up blended panorama emphasizes the need for photometric correction. Though naïve stitching provides us with a decent geometric registration, the exposure variation results in obvious seams. The latter pipeline is able to resolves these problems effectively, resulting in a visually consistent panorama.

Table 1: Comparison between Naive and Final Panorama

Aspect	Naïve Panorama	Final Panorama
Geometric Alignment	Good	Good
Visible Seams	Present	Largely Removed
Exposure Consistency	Poor	Uniform
Visual Quality	Moderate	High

8. Limitations

The method is developed with the assumption of constrained parallax among the images. Ghosting phenomenon may occur if there exists substantial camera translation or moving objects in the overlap regions. Also, the method requires enough texture for confident feature detection.

9. Conclusion

This project shows the process of a full-feature-based panorama stitching pipeline based on basic image processing technologies. With strong homography estimation, hand-held manual image warping together with exposure alignment and seamless blending, high quality panorama images can be generated. Comparison of naive and blended results highlights the necessity for photometric correction in panorama stitching.